Revisiting Bounded Lyapunov-Based Controllers

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1 Introduction

In my first foray into bounded Lyapunov-based controllers, I was not able to form the set Ω because Ψ seemingly had a discontinuity right at the origin. This made zero sense to me, but at the time I had no clue what was wrong; I had went through the calculations numerous times and verified that what I was doing seemed to be correct—the Lie derivatives were correct, the α and β expressions were correct, it all seemed fine, but for some reason the results were showing that there was a region extending from the origin that did not meet the stability criterion of $\dot{V} \leq -\lambda V$ for any value of λ.

I shelved that project for a while, but now that I'm getting closer to the end of my degree and I'm taking a class that will allow me to work on this project again, I chose to dig deeper into why my old methodology was incorrect. This time I think I got it; the equations are working, the system is behaving as I think it should, and I think I'm finally moving forward on this project. Read more to see what I came up with.

2 Bounded Controllers

The dynamics of bounded controllers is quite interesting. By our intuitive reasoning, we can say that there must be a region surrounding the equilibrium point where, outside of which, the controller cannot drive the states back to the equilibrium. This is the limit of our controller, and we must stay within this region. How can we obtain this region? There are two perspectives I can think of:

1. The steady-state, state-space perspective,
2. The optimal control, Lyapunov stability perspective.

2.1 Steady-State State-Space Perspective

Perhaps the most intuitive way to find the boundary of this bounded stability region is to use the state-space model of the system,

$$\dot{x} = Ax + Bu_{max} \tag{1}$$

where u_{max} is the maximum control input for a bounded controller. We don't need to worry about solving any complicated equations, we just need to set $\dot{x} = 0$ and find the state x that yields,

$$Ax + Bu_{max} = 0.$$

Solving this equation obtains the bounds of the region where our controller is stable in the Sense of Lyapunov (the state is not at the equilibrium but is bounded within a region of attraction). This solution alone, however, is limited because it does not show exactly *how* stable different states are within this region of attraction. We are certain that within this region we won't blow up, sure, but there is no way for us to tell how quickly we can expect our states to go towards the origin, if at all. Especially since we are using an implicit controller, we need more powerful theory to back us up.

2.2 Optimal Control, Lyapunov Stability Perspective

Enter the optimal control, Lyapunov stability perspective. This perspective has nonlinear system stability and optimal control theory behind it, allowing us to study more complex systems. For this, we need to introduce the control Lyapunov function (CLF),

$$V(x) = x^\top P x, \tag{2}$$

where P is a positive-definite symmetric matrix. If P is not positive-definite then the system is not globally asymptotically stable, and you either need to modify the way you calculate P or use a different CLF. Typically we would find P by solving the continuous-time Lyapunov equation,

$$A^\top P + PA + Q = 0, \tag{3}$$

which for any $Q \succ 0$ there exists a $P \succ 0$ that satisfies this equation. However, in the theory behind bounded control specified by Prof. El-Farra's own PhD work [1], we must use the continuous-time algebraic Riccati equation (CARE),

$$A^\top P + PA - PBR^{-1}B^\top P + Q = 0, \tag{4}$$

where $R \succ$ and $Q \succeq 0$. We will set $R = I$, and we can play around with Q to obtain different behavior. The CARE can be derived in the context of optimal control, where $Q \in \mathbb{R}^n$ is the state cost matrix and $R \in \mathbb{R}^m$ is the input cost matrix. R must be positive definite, otherwise there would be no control input to optimize. Q, however, is only required to be positive semi-definite, allowing us to set it to the zero matrix.

If we view the CARE from the perspective of optimal control, then by setting $R \succ 0$ we are saying we need to find some control input that minimizes the input cost over the horizon, and by letting $Q \to 0$ then we are saying we don't care where the states end up. In this case, as long as the states don't blow up, then we only need to worry about the control input (which is implicitly dependent on the state).

When $R = I$ and $Q \to 0$ we have,

$$\begin{aligned} A^\top P + PA - PBB^\top P &= 0 \\ P\left(2A - BB^\top P\right) &= 0, \end{aligned} \tag{5}$$

which requires either $P = 0$ or $P = 2\left(BB^\top\right)^{-1} A$, so we have two solutions (which makes sense, since the CARE is quadratic). The only sure solution we have is $P = 0$, since the other solution could blow up to infinity, especially if the eigenvalues of $\left(BB^\top\right)^{-1}$ are exceedingly large.

Realizing this in an actual simulation actually requires P to be just slightly larger than zero, like on the order of 10^{-7}, because $P = 0$ makes some of the analysis impossible (which is also why we need $Q \to 0$ and not $Q = 0$). However, a small P indicates that the system is close to being exactly balanced by the dynamics

of A and the inputs of B.

We need a little bit more theory, though. A valid Lyapunov function candidate is strictly positive for $x \neq 0$ and zero when $x = 0$, and it can be intuitively understood as the system attaining "energy" away from the origin; dynamical systems seek to minimize their energy, so the time derivative of the Lyapunov function must be negative:

$$\dot{V}(x) = \frac{\partial V(x)}{\partial x} \cdot \dot{x} \leq 0 \tag{6}$$

One way we can find where the time derivative is negative is by taking the Lie derivative of the Lyapunov function with respect to Ax and B—as a reminder, the Lie derivative tells us how much one vector field is changing as you move along another vector field. Finding the Lie derivatives of the Lyapunov function as it moves along Ax and B will tell us where the Lyapunov function is guaranteed to decrease in time, satisfying equation (6). The Lie derivatives of the LTI system, which I found in the previous document about bounded controllers, are,

$$L_f V = x^\top (2PA)x \quad \text{and} \quad L_G V = x^\top (2PB). \tag{7}$$

We now define the set Ψ such that a system with a bounded controller will remain trapped inside if the initial condition lies within it:

$$\Psi \doteq \left\{ x \in \mathbb{R}^n : L_f V + \lambda V \leq u_{max} \| \theta^\top L_G V^\top \| \right\} \tag{8}$$

$$\doteq \left\{ x \in \mathbb{R}^n : x^\top \left(2PA + \lambda P \right) x \leq u_{max} \| 2\theta^\top B^\top P \| \right\} \tag{9}$$

We can also define the set Ω such that a system with a bounded controller will have its Lyapunov function decay at a rate proportional to Λ if the initial condition lies within it:

$$\Omega \doteq \{ x \in \Psi : V(x) \leq c \} \tag{10}$$

$$\doteq \left\{ x \in \Psi : x^\top P x \leq c \right\} \tag{11}$$

It's okay if we are within Ψ, but we really want to be within Ω, because we can say for certain that our system has some guaranteed level of stability. The proof showing that Ψ and Ω have these properties is quite complicated and I understand very little of it, but I think I've come up with a more intuitive approach to proving these properties.

Proof. We begin by considering the continuous-time, LTI state-space representation of our system with multiplicative actuator fault θ, where the state matrix is modified such that the eigenvalues are shifted by a scalar value δ,

$$\dot{x} = (A + \delta I)x + B\theta u.$$

A controller designed around such a modified system will be inherently more robust when controlling the nominal system as the modified system has eigenvalues that lead to a less stable closed-loop system. Given a bounded controller $\|u\| \leq u_{max}$, we define the set Ψ such that any initial condition that lies within its interior yields a state-space model with $\dot{x} < 0$, $\forall x \in \text{int}(\Psi)$ and any intitial condition that lies on its boundary yields a state space model with $\dot{x} = 0$, $\forall x \in \partial_{\mathbb{R}} \Psi$:

$$\Psi \doteq \{ x \in \mathbb{R} : \exists \|u\| \leq u_{max} \text{ s.t. } (A + \delta I)x + B\theta u \leq 0 \}$$

We could alternatively define Ψ as the region in state-space where the time derivative of the Lyapunov

function is strictly negative for $x \notin \partial_{\mathbb{R}}\Psi$ and zero when $x \in \partial_{\mathbb{R}}\Psi$. The time derivative of the Lyapunov function of the modified state-space model is,

$$\dot{V}_{mod}(x) = x^\top(2PA + 2\delta P)x + x^\top(2PB\theta)u,$$

where the condition for it being strictly negative within the interior of Ψ and zero at the boundary allows us to state,

$$\Psi \doteq \left\{x \in \mathbb{R}^n : x^\top(2PA + 2\delta P)x \leq u_{max}\|2\theta^\top B^\top P\|\right\}.$$

We can finally define the constant $\lambda = 2\delta$ to obtain the original definition of Ψ:

$$\Psi \doteq \left\{x \in \mathbb{R}^n : x^\top\left(2PA + \lambda P\right)x \leq u_{max}\|2\theta^\top B^\top P\|\right\}$$

We notice, however, that we can split the term on the left side of the inequality in the set definition and rearrange to obtain,

$$x^\top(2PA)x + u_{max}\|2\theta^\top B^\top P\| \leq -\lambda x^\top Px,$$

where we recognize that the left side of the inequality is the Lyapunov function derivative for the nominal state-space model, $\dot{V}_{nom}(x)$, and the right side is equivalent to $-\lambda V(x)$, so we obtain,

$$\dot{V}_{nom}(x) \leq -\lambda V(x),$$

which is a property held only by Ω, by definition. We can also, for certain, say that $x^\top(2PA)x$ has one solution at the origin, so we can conclude that there are no other regions in state space where this inequality can "break". We can thus conclude that, for a linear LTI state-space model, $\Psi = \Omega$. □

From this, we hope to see that for a linear system, placing the initial conditions within Ψ but outside of Ω as it is defined in [1] and [2], it should make no difference on the decay rate of the Lyapunov function.

2.3 Adding a State Fault/Uncertainty, θ_a

For my project, I wanted there to be some fault/uncertainty on the state. This will allow me to explore FTC in numerous ways, not only through actuator faults. To do this, my system is as follows:

$$\dot{x} = A_c\theta_a x + B_c\theta_b u \tag{12}$$

There's no model system yet, I just wanted to make sure that the nominal model worked out before I made it any more complicated. For ease of typing, I'll define $\bar{A}_c = A_c\theta_a$ and $\bar{B}_c = B_c\theta_b$. So we have,

$$\dot{x} = \bar{A}_c x + \bar{B}_c u.$$

Using these matrices, then the Riccati equation becomes,

$$\bar{A}_c^\top \bar{P} + \bar{P}\bar{A}_c - \bar{P}\bar{B}_c\bar{B}_c^\top\bar{P} + Q = 0, \tag{13}$$

the Lyapunov function is,

$$V(x) = x^\top\bar{P}x, \tag{14}$$

and the time derivative of the Lyapunov function is,

$$\dot{V}(x) = x^\top(2\bar{P}\bar{A}_c)x + x^\top(2\bar{P}\bar{B}_c)u. \tag{15}$$

Ψ is now defined as,

$$\Psi \doteq \left\{x \in \mathbb{R}^n : x^\top(2\bar{P}\bar{A}_c + \lambda\bar{P})x \leq u_{max}\|x^\top(2P\bar{B}_c\|\right\}, \tag{16}$$

and as the proof above showed, there is no need to define Ω once we've defined Ψ, as all system trajectories within Ψ will have their associated Lyapunov function decay at a known rate in time. To finish it off, for modeling with MPC, we need to discretize the state-space model over a sampling interval Δ,

$$A = e^{\bar{A}_c\Delta} \quad \text{and} \quad B = \bar{A}_c^{-1}\left(e^{\bar{A}_c\Delta} - I\right)\bar{B}_c, \tag{17}$$

which gives us the discrete-time state-space model,

$$x_{k+1} = Ax_k + Bu_k. \tag{18}$$

3 Results

To start us off on the results, I modeled an MPC-controlled system with some input constraints to verify that my simulation does indeed provide accurate results. These are seen in figure (1). What's interesting that we can already see in this figure is that to controller only really activates when the unstable state (in this case, the temperature) is away from its equilibrium point. The controller is able to account for the fact that it only really needs to stabilize the temperature, and once that's done then the concentration will naturally equilibrate itself.

Next, using the definition of Ψ in equation (9), we can find the points within state space that are contained within Ψ. These are seen in figure (2). The green portion is all the points that are contained within Ψ, and all the white is outside of Ψ. What I wanted to investigate was what sort of Ψ do we see when we set $\lambda = 0$? Meaning, when $\dot{V}(x) = 0$, we expect that at states located at the boundary of Ψ, our bounded controller must have the same "strength" to tug the states towards the equilibrium point that the system has to pull itself away—at the boundary of $\Psi(\lambda = 0)$, we expect to see a system that is only stable in the sense of Lyapunov (or perhaps not even Lyapunov stable?).

To find where the critical state, x_{crit}, is, we can use a solver in two ways. The first way is perhaps the most straightforward way: through the state-space model. We know that to get the derivative of the states to be zero (meaning that the time derivative of the Lyapunov function is also zero), then we can solve for the state where $\bar{A}_c x_{crit} + \bar{B}_c u_{max} = 0$. Since concentration is the stable state, I specified a concentration and had `scipy.optimize.fsolve` find the temperature that made the derivative zero.

Another way we can use a solver is to set $\lambda = 0$ and $Q = 0$, which by definition will make $\dot{V}(x) = 0$. And by setting $Q = 0$, we are saying that we don't care about where the states end up, just calculate control input that minimizes the cost. Since at the boundary where $\lambda = 0$, the minimum cost would be where $\|u\| = u_{max}$, as any larger control input is not possible and any smaller would cause the states to violate their respective constraints, since they would grow without bound. We essentially need to find where $x^\top(2\bar{P}\bar{A}_c + \lambda\bar{P})x - u_{max}\|x^\top(2\bar{P}\bar{B}_c)\| = 0$.

Both solutions yield the same critical point, within computational error. For example, the first method (solving where the state-space model has a zero derivative), at $C_{crit} = -0.1$, it finds $T_{crit} = 11.909926777353618$. The second method at the same C_{crit} finds $T_{crit} = 11.909926778476086$, off by literally less than 9.5×10^{-9}. This confirms any suspicion that the two solutions are not equivalent.

When we place the initial conditions near x_{crit}, we see that the controller is barely able to steer it towards the equilibrium in figure (3). When we place the initial condition exactly equal to the critical condition, we see that the controller is able to stabilize the system to a critical degree in figure (4).

We can also see the affect of adding actuator faults through θ_b on the size of Ψ. When we add faults of magnitudes $\theta_b \in [1, 0.8, 0.5]$, we see the shape of Ψ get smaller around the equilibrium point. Ellipses have been drawn to form Ω for each fault through the original definition of Ω. This is seen in figure (5).

Since, however, we found that $\Omega = \Psi$, several iterations of the simulation were started with the initial conditions both inside and outside Ω to see if any of them resulted in a violation of $\dot{V}(x) \leq -\lambda V(x)$. Some of these simulations are seen in figure (6).

From a state-space perspective, we can see that all but one of these trajectories lie outside Ω. We can check the time derivative of each trajectories Lyapunov function as the simulation progresses, which are seen in figure (8). We see that all of the trajectories satisfy the condition $\dot{V}(x) \leq -\lambda V(x)$. This experimentally confirms, at least, that there exist some states outside Ω that satisfy the time derivative condition. However, I would still like to check that my proof that $\Omega = \Psi$ is correct. I feel like it would need more, but I can check with Prof. El-Farra.

4 Conclusion

This study confirms that the sets defined by El-Farra [1] and Xue [2] accurately describe the region where a closed-loop system controlled via a bounded controller remains stable. I feel like I've made some strides here that would make great additions to my thesis.

References

[1] N. H. El-Farra and P. D. Christofides, *Control of nonlinear and hybrid process systems: Designs for uncertainty, constraints and time-delays*. Springer, 2005.

[2] D. Xue and N. El-Farra, "Forecast-triggered model predictive control of constrained nonlinear processes with control actuator faults," *Mathematics*, vol. 6, p. 104, Jun 2018.

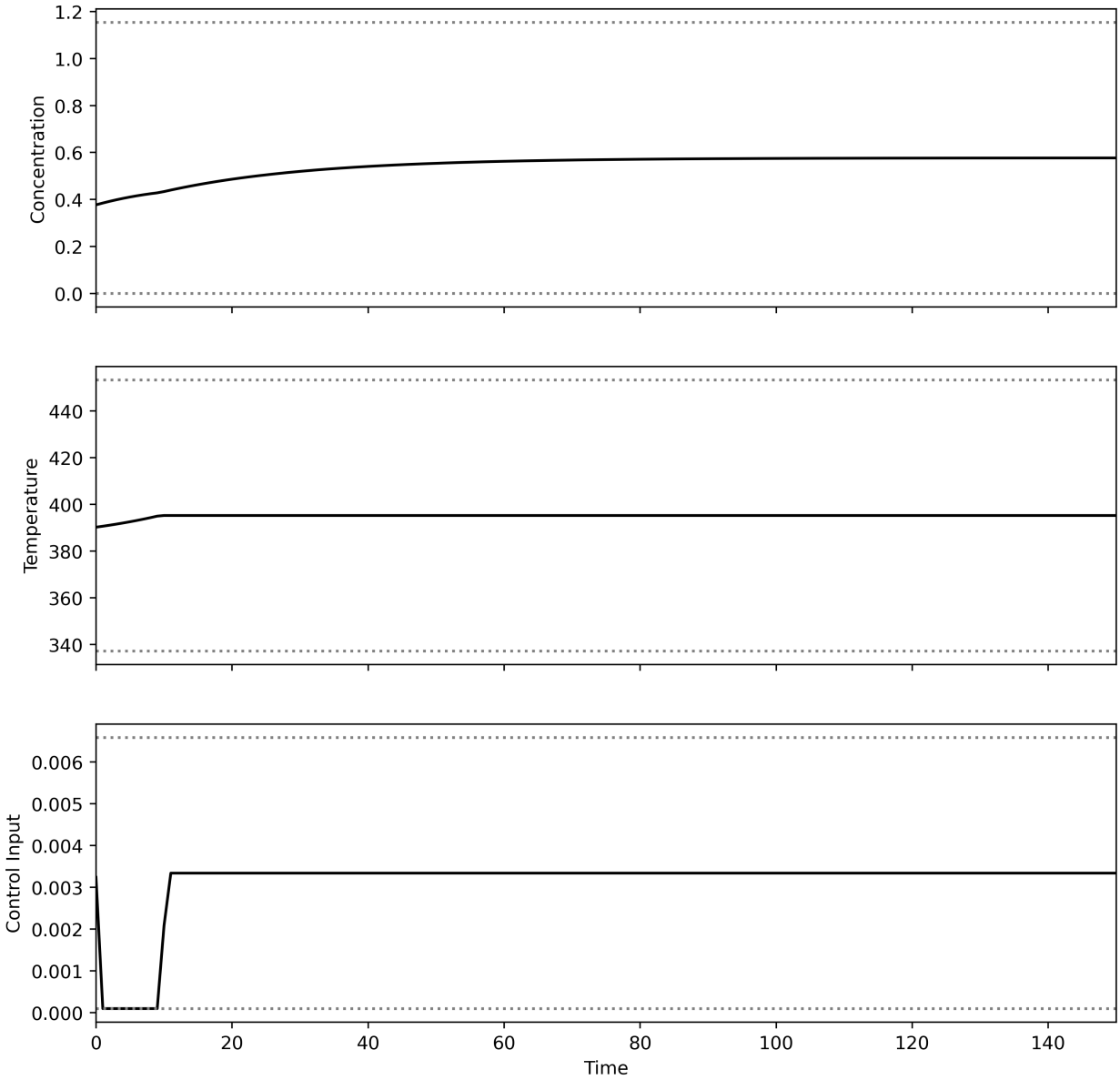


Figure 1: The dynamic response of the system at an arbitrary initial condition away from the equilibrium. MPC is doing its job by staying within the input constraints. Temperature is the unstable one of the two states, and we notice that once temperature has stabilized, the controller stops activating and concentration, the stable of the two states, creeps back to the equilibrium point on its own.

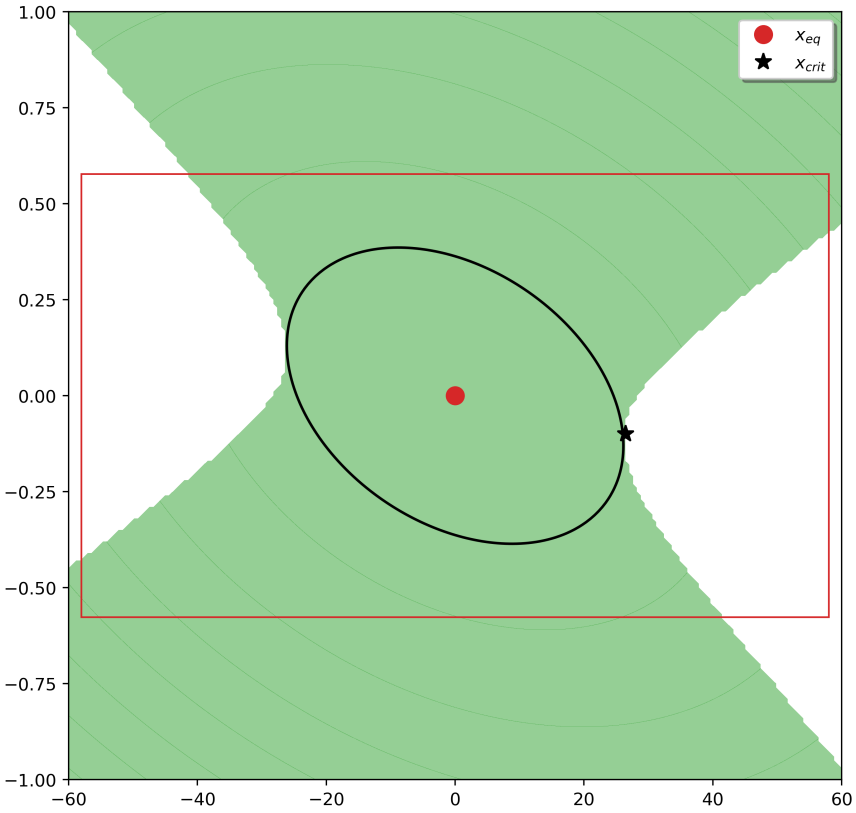


Figure 2: The set Ψ (shaded green) with equilibrium point (red dot). The black circle is the set Ω using the original definition, and the black star is an initial condition chosen such that the controller is marginally able to stabilize the states. $\lambda = 0$. The red rectangle is the polytope of state constraints, $\mathcal{X}$. It isn't really of use to us right now, but I plan to use it in my project.

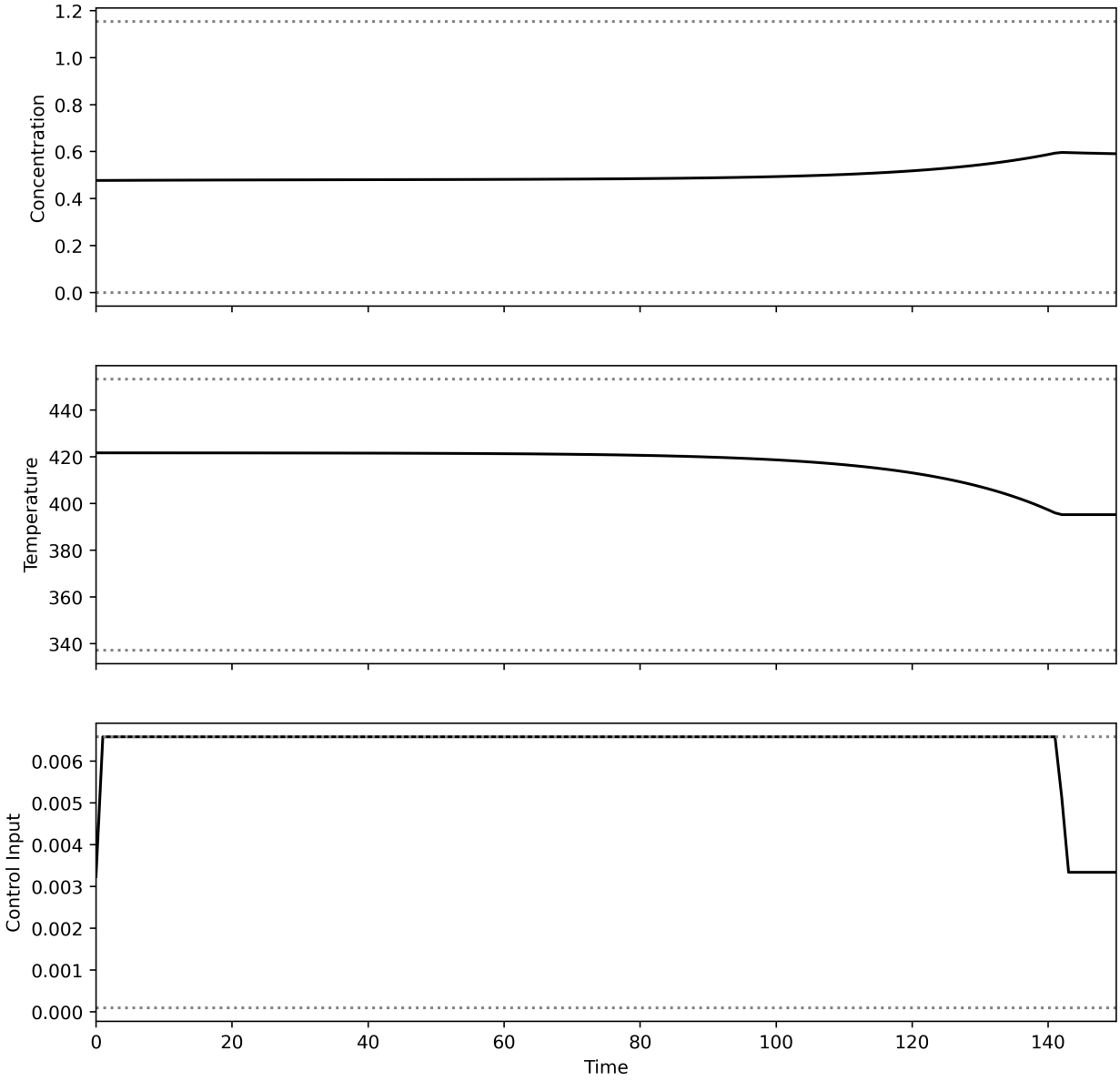


Figure 3: The dynamic response of the MPC-controlled system starting at an approximately critical initial condition found in figure (2). We can see that the critical state does indeed yield a system that is just barely able to steer the states away from instability. When $\lambda = 0$ and $Q \to 0$, the boundary of Ψ yields the states at which the controller and unstable system have an equal "tugging force," canceling each other out and yielding a system that is stable in the sense of Lyapunov.

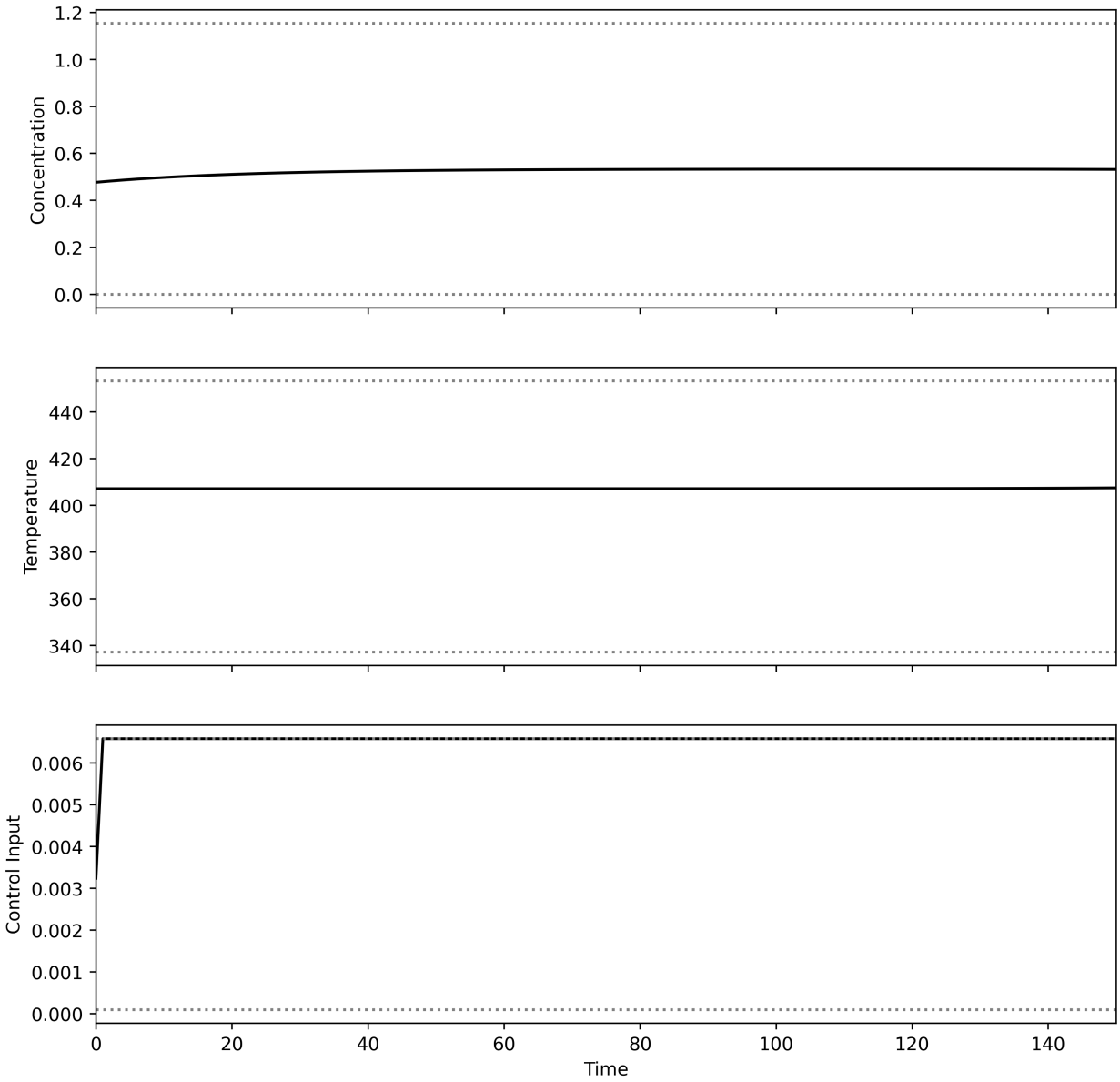


Figure 4: We see that, placed at exactly the critical initial condition, the system is able to stabilize the system to a critical degree, meaning that there is literally no wiggle room for the controller to back off even the slightest bit. The bounded controller and the unstable system perfectly cancel each other out.

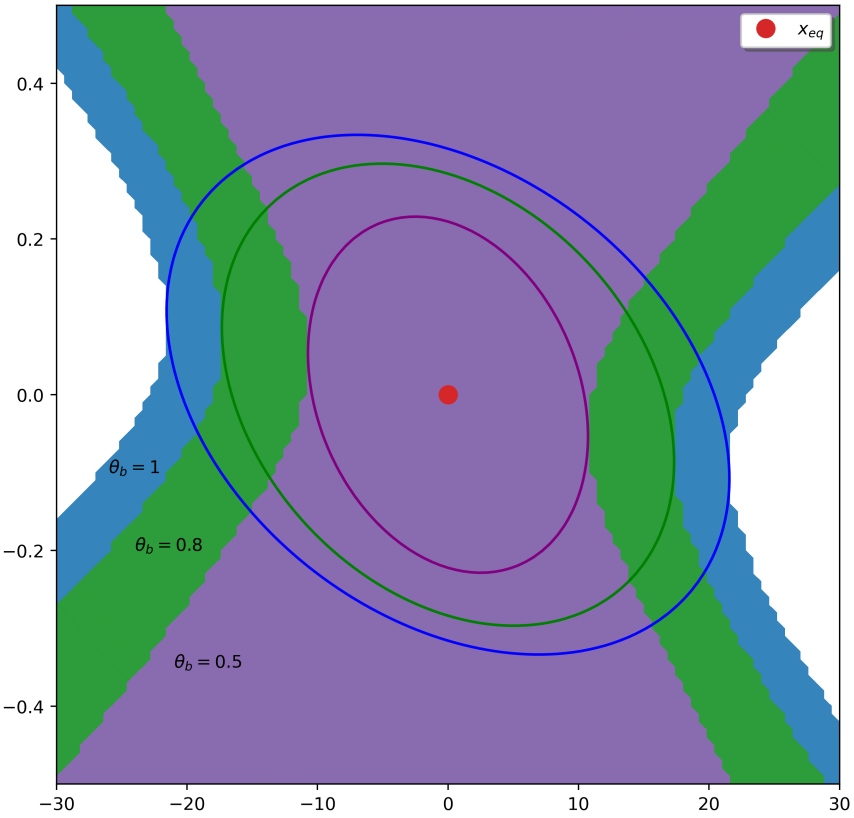


Figure 5: When we impose actuator faults of magnitude $\theta_b \in [1, 0.8, 0.5]$ and $\lambda = 0.01$, we can see that as the faults grow larger, Ψ shrinks. As Ψ shrinks, so too should Ω.

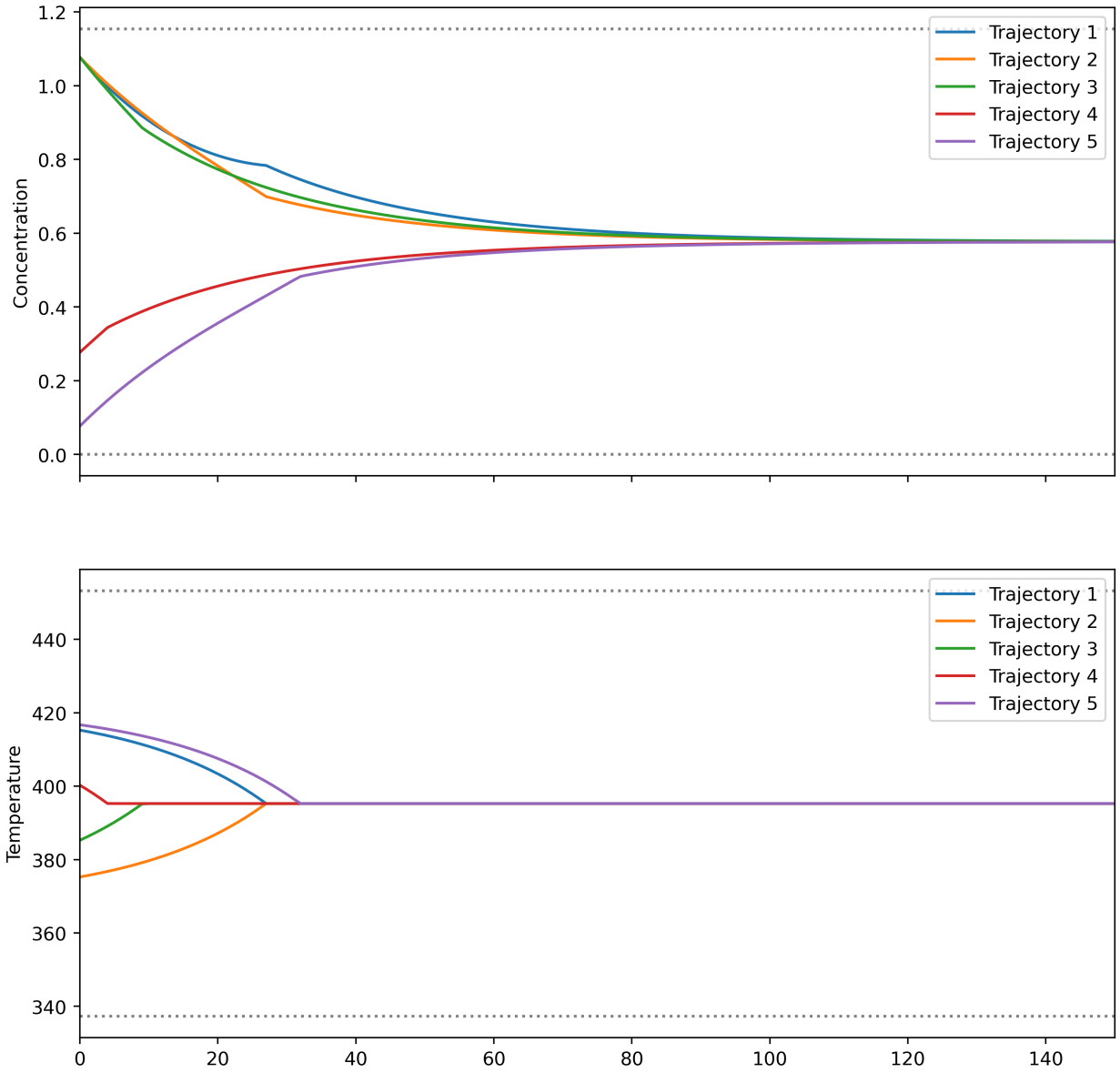


Figure 6: Multiple initial conditions were tested, many of which outside of Ω using the original definition, to see whether or not they are all stable and satisfy the condition that $\dot{V}(x) \leq -\lambda V(x)$. At least from this plot we can see that the controller is indeed able to stabilize the system.

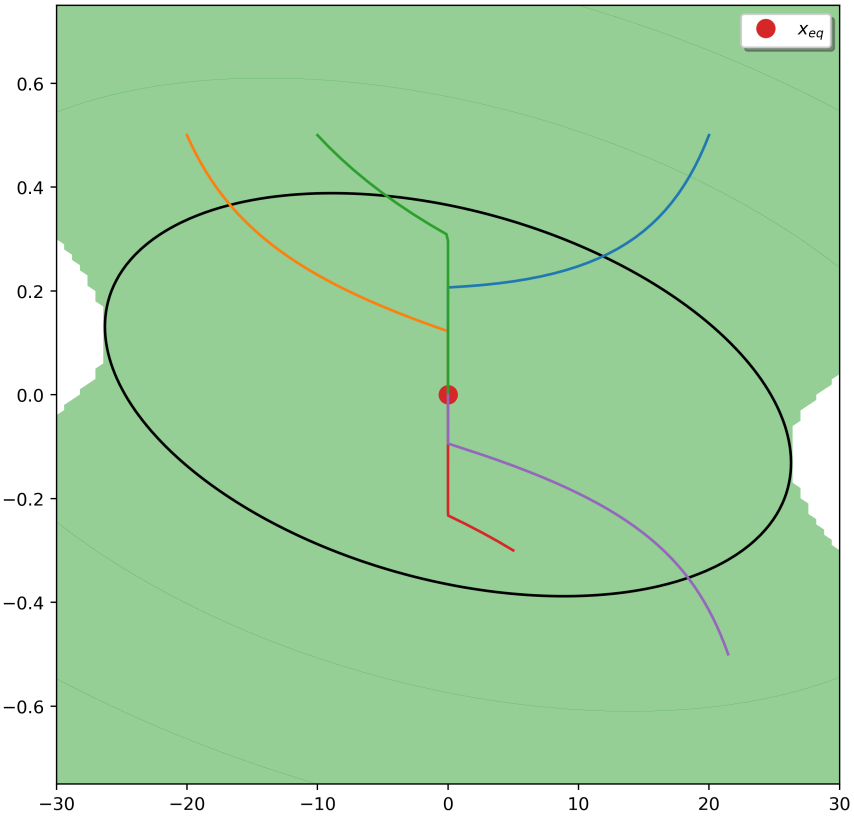


Figure 7: The state-space view of the trajectories in figure (6) shows that only one of them was chosen to be within Ω, all others were specifically chosen to be inside Ψ and outside Ω. From this state-space view we can see that the system inherently "aims" the unstable states (in this case, the temperature, x-axis) towards their respective equilibrium point, and once there the stable state (in this case, the concentration, y-axis) can settle down to the origin on its own. Once the unstable state has settled, we actually see in dynamic simulations that the controller doesn't move away from its own equilibrium.

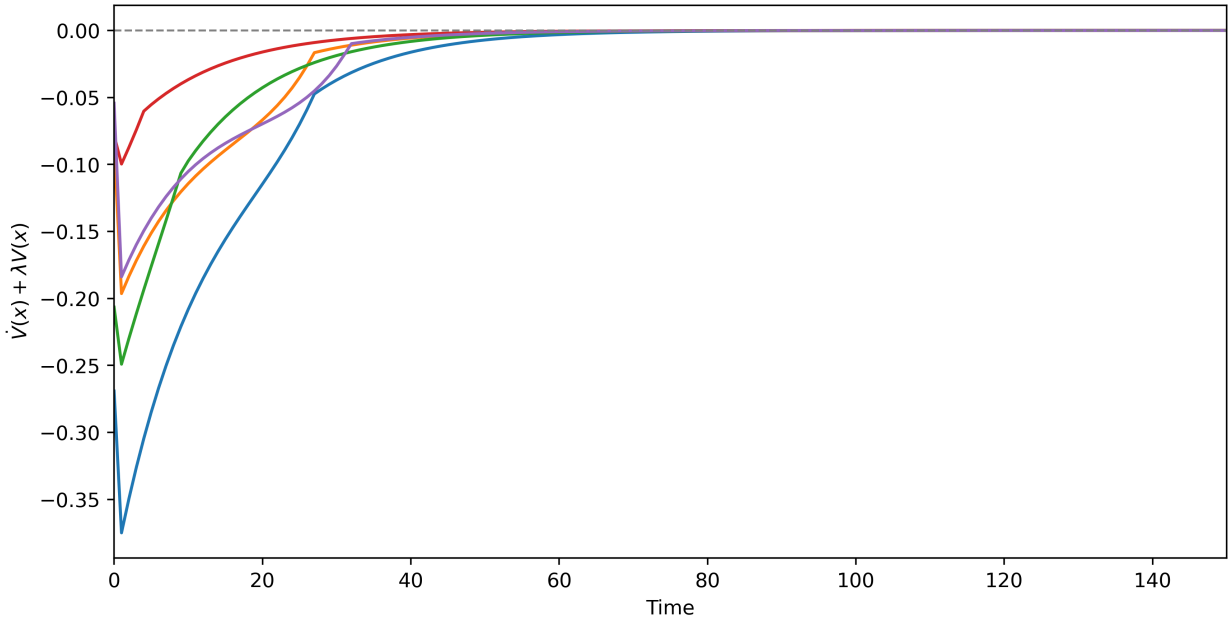


Figure 8: The plot of $\dot{V}(x) + \lambda V(x)$ shows that, regardless of where the states start, the time derivative of the Lyapunov function will always decay at a rate less than or equal to $-\lambda V(x)$. This example shows that for linear systems, Ψ and Ω are identical sets.